

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T15 · Direct & Inverse Proportion

Questions

Student book · questions with writing space

■ 12 pages

FAMILY MAP

Direct Inverse Proportion

F01

Build an inverse-variation model

8 Q | 25 marks

F02

Build an algebraic direct-variation model

3 Q | 12 marks

(Total for Question 18 is 4 marks)

Question 22

4MA1-2H Q17

MODULAR UNIT 2

3 marks

17 Q is directly proportional to the square root of d

$$Q = 4.5 \text{ when } d = 324$$

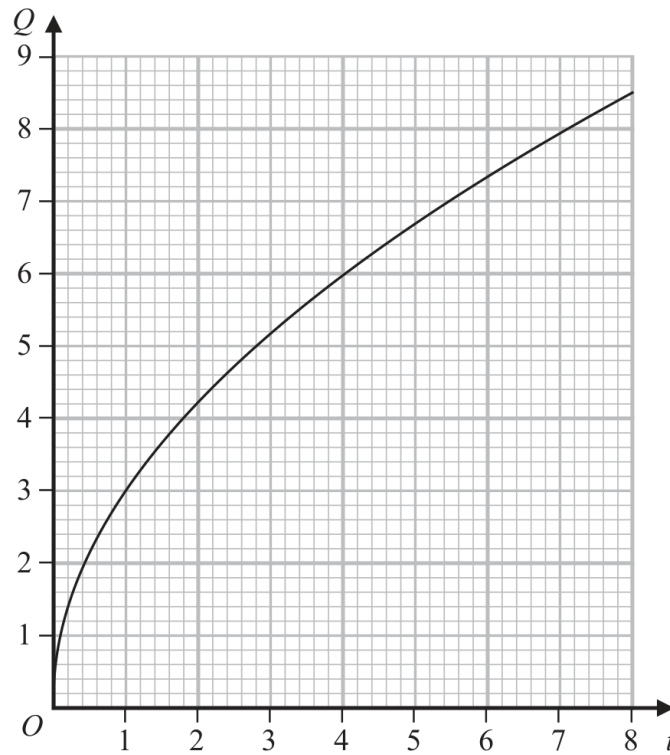
Find a formula for Q in terms of d

Question 26

4MA1-2H Q16

MODULAR UNIT 2

5 marks

16 Q is directly proportional to $\sqrt{t}$ The graph shows the relationship between Q and t for $0 < t < 8$ (a) Find a formula for Q in terms of t

(3)

 Q is increased by 20%(b) Find the percentage increase in t

Question 11

4MA1-1H Q14 M01

MODULAR UNIT 2

3 marks

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

.....

4MA1-2H Q17 Q17

4 marks

- Find the value of h when $M = 500$

Question 10

4MA1-1H Q15

MODULAR UNIT 2

3 marks

15 A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C = \dots\dots\dots$

$A = 5$ when $r = 0.3$

(3)

(b) Find the value of A when $r = 7.5A$

$A = 5$ when $r = 0.3$

(3)

(b) Find the value of A when $r = 7.5A$

14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

.....
(Total for Question 14 is 3 marks)

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c

Give your answer in its simplest form.

(Total for Question 17 is 3 marks)

17 T is inversely proportional to $\sqrt{m}$

$$T = 15 \text{ when } m = 36$$

Find a formula for T in terms of m

(Total for Question 17 is 3 marks)